RR-W-670D
October 17, 1972
SUPERSEDING
Fed. Spec. RR-W-670C
June 16, 1959

FEDERAL SPECIFICATION

WRINGER, MOP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 *Scope.* This specification covers pressure-plate squeeze-type wringers and gear and rack downward pressure wringers generally used with buckets during wet mopping operations. Mop wringers and bucket combinations are covered by RR-M-1075. Buckets are covered by RR-B-800.

1.2 Classification.

1.2.1 Types, classes, and sizes. Wringers shall be of the following types, classes, and sizes, as specified (see table I and 6.2).

Type I - Pressure-plate squeeze-type wringers.

Class 1 - Horizontal pressure squeeze.

Size 1 - 7-1/2 x 6 x 5 inches.
Size 2 - 7 x 7-1/2 x 5-3/4 inches.
Size 3 - 7-1/2 x 8 x 6-1/2 inches.

Class 2 - Combination horizontal and downward pressure squeeze.

Size 2 - 7 x 7-1/2 x 5-3/4
Size 4 - 8-1/2 x 9 x 6-3/4 inches.

Type II - Gear and rack wringers.

Size 1 - 7-1/8 x 3-1/2 x 4-5/8 inches.
Size 2 - 7-7/8 x 5-3/8 x 5-1/4 inches
Size 3 - 8-7/8 x 5-1/4 x 6-1/8 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

T-M-561 - Mophead, Wet, Cotton or Cotton and Rayon.
RR-B-800 - Bucket, Mop (Metal).
RR-M-1075 - Mopping Outfits, Floor.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-C-650 - Crates, Wood, Open and Covered.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 2240-68 - Indentation Hardness of Rubber and Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Bid samples. When specified, bid samples of wringers of each type, class, and size required (see 1.2.1) shall be submitted for examination and testing. The bid samples shall be representative of the wringer the supplier proposes to furnish under contract (see 4.2).

3.2 Materials.

3.2.1 Metal. All metal parts of wringers shall be steel or malleable iron, electroplated to resist corrosion under normal service conditions except as otherwise specified herein.

3.2.2 Plastic. Handle grips shall be made of a suitable plastic material such as polyvinyl chloride, having a Shore A hardness of 75 to 80 when determined as specified in ASTM D 2240-68. The material shall be nonmarking.

3.3 Design. Wringers shall be designed as follows:

(a) To fit over the side of a bucket of the recommended type and size (see 6.3) and operate without hanging up or permanently damaging the wringer or bucket to the extent that serviceability or durability is impaired.
(b) To provide for draining surplus water from mops into a water container of proper size (see 6.3).
(c) To open when the handle is brought to a vertical position and close automatically when the handle is moved to a horizontal position.
(d) To have proper weight distribution so that when in place on a bucket of proper size (see 6.3) stability of the container will not be affected.

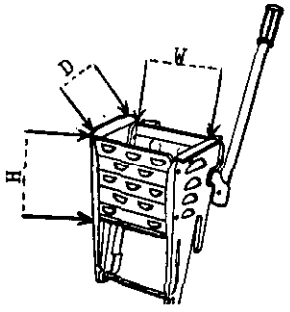
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3.4 Construction.

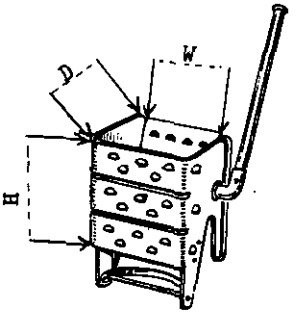
3.4.1 Type I - Pressure-plate squeeze wringers.

3.4.1.1 Class 1 - horizontal pressure squeeze. The wringer shall be similar to figure 1A. Wringer components shall be made of metal as specified in 3.2.1, except that covers for the wringer mechanism and name plates may be coated with baked-on enamel. The mop compartment shall have perforations or louvers designed and located so that water from wet mops will drain into the bucket to be used with the wringer (see 6.3). When the handle is in approximately vertical position, the mop compartment shall be open as shown in figure 1A or 1B. When pressure is applied to the handle and the handle is moved from vertical to approximately horizontal position, the moveable pressure plate at the back of the compartment shall move forward toward the opposite side of the compartment to squeeze the mop against it. When pressure is released, the handle shall return to vertical position opening the mop compartment to its full capacity (see table I). The lower portion of the wringer extending below the mop compartment shall be designed to fit over the side of the bucket (see 6.3) so that the wringer will operate as specified in 3.3(a).

3.4.1.2 Class 2 - Horizontal and downward pressure squeeze. The wringer shall be similar to figure 1B and constructed as specified in 3.4.1.1, except that the motion of the pressure plates shall be horizontal and downward.



A. Class 1 - Horizontal pressure wringer



B. Class 2 - Combination horizontal and downward pressure wringer

W = width (side to side)
D = depth (front to back)
H = height (top to bottom

FIGURE I. PRESSURE-PLATE SQUEEZE WRINGERS

3.4.2 Type II - Gear and rack wringers. Gear and rack wringers shall be similar to figure 2. Wringer components shall be made of metal as specified in 3.2.1, except that gear covers and name plates may have a baked enamel finish. The mop compartment shall have perforations or louvers as shown in figure 2, designed and located so that water from wet mops will drain into the bucket to be used with the wringer (see 6.3). Gears shall be rack and pinion type with spur design, formed by welding the two units together, with gear teeth offset so that it is impossible for them to slip out of mesh either horizontally or vertically. Gears shall be completely enclosed to prevent entanglement of mop strings. To reduce wear, a steel bushing shall be provided at each end of the gear shaft where the shaft enters each side plate. When the handle is in approximately vertical position, the pressure bars shall rest at the sides of the wringer compartment in vertical position. When pressure is applied to the handle, the pressure bars shall fold over from vertical position to horizontal position parallel to the bottom of the wringer and travel downward as the handle is brought to approximately horizontal position to squeeze the mop against the bottom of the compartment. When pressure is released, the handle shall automatically return to vertical position opening the mop compartment to its full capacity (see table I). The pressure plates shall be secured in the guide slots at the sides of the wringer by solid journals with ends peened over or tapped and driven. The lower portion of the wringer extending below the mop compartment shall be designed to fit over the side of the bucket (see 6.3) so that the wringer will operate as specified in 3.4(a).

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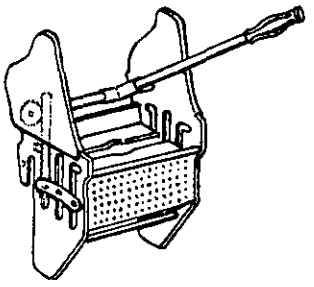


FIGURE 2. GEAR AND RACK WRINGER

3.4.3 Handles. Handles of all wringers shall be steel tubing or pipe fitted at one end with a handle grip. The handle shall be coated to resist corrosion by electroplating or by a baked enamel finish. The handle shall fit snugly into the socket without slack and shall function without locking in an open or closed position when subjected to abrupt application or release of load.

3.5 Dimensions.

3.5.1 Type I wringers. When determined at the points shown in figures 1A and 1B as specified in 4.3.2.3.1, the outside dimensions of the mop compartment of type I pressure-plate squeeze wringers shall be in compliance with the applicable requirements specified in table I.

3.5.2 Type II wringers. When determined as specified in 4.3.2.3.2, the dimensions of gear and rack mop compartments shall be in compliance with the applicable requirements specified in table I.

3.5.3 Handles. When determined as specified in 4.3.2.3.3, the dimensions of handles shall be in compliance with the applicable requirements in table I.

TABLE I. Dimensions of wringers

	Type I					Type II		
	Class 1			Class 2				
	Size 1	Size 2	Size 3	Size 2	Size 4	Size 1	Size 2	Size 3
Mop compartment:								
Width (side to side), minimum..... inches	7-1/2	7	7-1/2	7	8-1/2	7-1/8	7-7/8	8-7/8
Height (top to bottom), minimum.....do.	6	7-1/2	8	7-1/2	9	3-1/2	5-3/8	5-1/4
Depth (front to back), minimum.....do.	5	5-3/4	6-1/2	5-3/4	6-3/4	4-5/8	5-1/4	6-1/8
Pressure plates or bars:								
Width, minimum..... inches	---	---	---	---	---	1-7/16	1-3/4	2
Distance from bottom of mop compartment when handle is in horizontal position, maximum..... inches	---	---	---	---	---	1-1/2	1-7/8	2-1/16
Handle:								
Length, minimum..... inches	12-1/2	13-1/2	16	13/1/2	16	12-1/2	13-1/2	16
Diameter, minimum..... do.	0.875	0.875	0.875	0.875	0.875	0.875	0.875	0.875
Wall thickness, minimum . do.	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0.075

3.6 Resistance to deformation. When tested as specified in 4.4.1 under a load of 250 ± 5 pounds, wringer parts, including handles, shall not loosen, break, separate, permanently deform, or be damaged in any way that will affect normal functioning of the wringer or cause the wringer to hang up on the side of the bucket.

3.7 Identification marking. Unless otherwise specified, wringers shall be permanently and legibly marked with the name or trademark easily identifiable with the manufacturer, or both.

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3.8 Workmanship. Wringers shall be free from sharp edges that might snag or cut the mop yarns or injure the operator; from oil, grease, or dirt; and from other defects that might affect serviceability or durability. Normal factory lubrication of gears, bushings, and other points of friction shall not be considered to be dirt. When rivets are used they shall be neatly set or welded. All welds shall be complete and smooth with no evidence of burned through areas, porosity, cracks, or molten metal spatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examinations and testing set forth in the specification where such are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Bid samples. Bid samples submitted in accordance with 3.1 shall be examined for visual defects listed in table II and for dimensional defects as specified in 4.3.2.3.1 through 4.3.2.3.3 and tested for resistance to deformation as specified in 4.4.1.

4.3 Sampling for inspection and acceptance.

4.3.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with the specified requirements of referenced subsidiary specifications and standards, or, if none, in accordance with this specification.

4.3.2 End item examination and testing.

4.3.2.1 Lot. For the purpose of examination and testing of wringers, a lot shall consist of all wringers of one type, class, and size offered for delivery at one time.

4.3.2.2 Visual examination. Wringers shall be examined visually for defects in table II. The inspection level shall be II and the acceptable quality level (AQL) 4.0 defects per 100 units. The unit for examination shall be one wringer. If the number of visual defects in the sample exceeds the acceptance number specified in MIL-STD-105, the entire lot of wringers shall be considered defective.

TABLE II. List of visual defects for wringers

Examination	Defects
Type, class, and size	Not as specified.
Design and construction:	
Finish	Not as specified; can be easily removed by scraping with knife blade.
Fit and balance	Wringer does not fit over side of bucket (see 6.3); weight not properly distributed so that bucket will remain stable when wringer is in place and during wringer operation.
Pressure bar journals (type II)	Ends not solid and peened over or drilled and tapped. Not provided with steel bushings as specified.
Gears (type II)	Not covered.
Grips for handle	Absent; not nonmarking.
Handle	Does not fit securely in socket without slack.
	When in approximately vertical position mop compartment is not open to full capacity.
	When in approximately horizontal position mop compartment is not closed.
	Handle locks when pressure is applied or released abruptly.
Operation	Wringer operation cannot be completed without damaging bucket or wringer to the extent that serviceability or durability would be adversely affected (see note a). Gears slip.
Workmanship	Rivets (when used) not neatly set or welded. Welds not complete and smooth; evidence of burned through areas.

a/ Wringer mechanism hangs up completely during operation because of failure to clear the side of bucket or because it does not clear the side of the bucket sufficiently to avoid damage to wringer or bucket that would impair its use. (Slight scratches that do not completely remove the finish are not considered to be defects.)

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4.3.2.3 Dimensional examination. Wringers shall be examined as specified in 4.3.2.3.1 through 4.3.2.3.3 for conformance with the dimensional requirements in table I. The inspection level shall be S-3 and the acceptable quality level (AQL) 6.5 expressed in terms of percent defective. The unit for examination shall be one wringer. If any wringer in the sample for examination fails to meet any dimensional requirement for which examination is made, it shall be considered defective. If the number of defective units in the sample exceeds the acceptance number specified in MIL-STD-105, the entire lot of wringers shall be considered as having failed to meet the requirements of the specification.

4.3.2.3.1 Type I - Pressure-plate squeeze-type wringers. The top-to-bottom, front-to-back, and side-to-side dimensions of the squeeze-type wringers shall be determined by measuring from edge to edge at the points shown on figures 1A and 1B, by means of a steel rule or other suitable measuring device.

4.3.2.3.2 Type II - gear and rack wringers. The following inside dimensions of mop compartments of gear and rack wringers shall be determined by means of a template or other suitable measuring device.

Height (top to bottom). Distance from the center bottom of the wringer along a line perpendicular to the bottom of the wringer, to a point where the pressure plates are level with the side of least height.

Width (side to side). The distance from the top of the side least in height to a point directly opposite on the other side.

Depth (front to back). The distance from the top of the side least in height to a point directly opposite on the other side.

Check each result for conformance with the applicable requirement in table I.

4.3.2.3.3 Dimensions of handle. (a) Length. Measure the length of the handle from the center of the pivoting position to the end of the hand gripping portion. (b) Diameter. Measure the inside and outside diameter of the handle. Subtract the inside diameter from the outside diameter to obtain the wall thickness of the handle. Check each value obtained for length of handle, inside diameter of handle, and wall thickness of handle for conformance with the applicable requirements in table I.

4.3.2.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 are complied with. Defects shall be scored in accordance with defects listed in table III. The lot shall be the number of containers offered for delivery at one time. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per one hundred units.

TABLE III. Examination of preparation for delivery

Examine	Defect
Markings (exterior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Inadequate application of components such as strapping and sealing, inadequate, improper or missing; components improperly secured; bulging or distortion of containers, pads or forms inadequate or missing.
Number	Number per container is more or less than specified.

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4.3.2.5 Testing. A random sample of wringers shall be taken from each lot in accordance with table IV and subjected to the test specified in 4.4.1.

Table IV. Sampling for lot acceptance tests

Number of wringers in lot	Number of wringers in sample	Acceptance number (defective)	Rejection number (defective)
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 800	15	1	2
801 to 1300	20	1	2
1301 and over	30	2	3

4.4 Test procedure.

4.4.1 Resistance of wringer to deformation. Place a block of wood or other suitable material in the mop compartment, of sufficient thickness so that the pressure plate or bars, as applicable, bottom on the block when the handle is in horizontal position. Apply a load of 250 $\pm$ 5 pounds to the handle where the hand normally grips and hold the load for 5 to 10 seconds. Examine the wringer for conformance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packing. Mop wringer outfits shall be packed level A, B, or C, as specified.

5.1.1 Level A. Ten mop wringers shall be nested in a manner to effect minimum cube and packed in wood-cleated plywood or nailed-wood boxes conforming to overseas requirements of Federal Specifications PPP-B-601, PPP-B-621, or in unsheathed wood crates conforming to PPP-C-650. Closure of boxes and crates shall be in accordance with the applicable specification.

5.1.2 Level B. Except where the handles are of the drive fit and/or rigidly fixed type, the handle of the wringer shall be removed and completely wrapped with kraft paper of not less than 50 pounds basis weight. Each wringer complete with wrapped handle shall be packed in a box conforming to PPP-B-636, class domestic, grade not less than 200. The box shall be fitted with either die-cut or folded interior forms of fiberboard to prevent excessive movement of the wringer within the box during shipment and handling. Boxes shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C. Mop wringers shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

6. NOTES

6.2 Ordering data. Purchasers should exercise any options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, and class of wringer required.
(c) When identification marking other than that specified in 3.7 is required.
(d) Applicable levels of packing and packaging.

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TABLE V. Recommended size combinations of wringers, buckets, and mops

Wringer sizes	Mop sizes[a] ounces	Bucket sizes[b] quarts
Type I		
Class 1		
Size 1	12-16	16
Size 2	16-20	26
Size 3	24-32	35
Class 2		
Size 2	12-20	26
Size 4	12-32	35
Type II		
Size 1	12-16	16
Size 2	16-24	26
Size 3	24-32	35

[a] Cotton mops described in Interim Federal Specification T-M-00561D, or equivalent.

[b] Type D (oval) buckets described in RR-B-800.

Civil Agencies Coordination Activities:

GSA - FSS, PBO

Preparing activity:

GSA - FSS

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